

In this Standard the comma has been adopted as the decimal sign.

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1. SCOPE

This Standard establishes the aerodynamic smoothness requirements for skin contour of ERJ 170 and ERJ 175 aircraft, such as steps or surface discontinuities, gaps, installation of fasteners, angular tolerances for control surfaces, slats and flaps, component angular tolerances and other configurations established on the production drawings.

In order to guarantee that the aerodynamic requirements are being correctly indicated in manufacturing and assembly drawings and appropriately monitored during aircraft manufacture process, the Embraer Aerodynamics Engineering Group is supposed to evaluate and approve the applicable inspection methods and test procedures that will be adopted to control all relevant characteristics that are defined in this Standard.

2. APPLICATION

This Standard is applicable to all external surfaces of ERJ 170 and ERJ 175 aircraft, unless otherwise specified on production drawings.

3. APPLICABLE DOCUMENTS

Embraer Standards

NE 03-004 - Surface Roughness.

NE 03-006 - Rivets and Pins - Drilling, Countersinking and Dimpling.

NE 03-068 - Bolts And Screws - Drilling, Countersinking and Dimpling for Installation.

NE 03-085 - Surface Roughness – Inspection.

NE 07-038 - EMB-145, EMB-314, EMB-390, EMB-500/505, EMB-550 and ERJ 170/190 Visual Defects - Rework and Acceptance Criteria for External Areas.

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4. SURFACE CLASSIFICATION

The aircraft external surfaces are divided into three areas (“IA”, “IB” and “II”), according to aerodynamic smoothness, to facilitate the manufacturing and inspection procedures. The definitions of the extent of each area are given as follows in Table 1. See Figure 1, Figure 2 and Figure 3.

TABLE 1 : CRITICAL, SEMI-CRITICAL AND NON-CRITICAL AREAS

AREAS			DESCRIPTION
CRITICAL	“IA”	Fuselage	– Radome.
		Nacelle	– From lip of the inlet cowl to 406,4 mm (16 in) aft (internal and external surfaces.
		Pylon Fairing	– The entire pylon fairing external surface.
		Wing	– From the wing-fuselage intersection up to station Y = 4600, from the leading edge up to the spar 3, upper and lower surfaces.
			– From station Y = 4600 to the wing tip, from the leading edge up to spar 2, upper and lower surfaces.
		Winglet	– From winglet root to tip.
			– From the leading edge up to aft spar, upper and lower surfaces.
		Wingtip	– From wingtip root to tip.
			– From the leading edge up to aft spar, upper and lower surfaces.
		Slats	– The entire outer surface of the retracted slats.
		Flaps	– From the leading edge up to 40% of the local chord for all flap panels.
SEMI-CRITICAL	“IB”	Horizontal Stabilizer	– From the leading edge up to front spar, upper and lower surfaces.
		Vertical Tail	– From the leading edge up to front spar.
		Fuselage	– Forward fuselage, except radome.
		Horizontal Stabilizer	– From front spar to trailing edge, upper and lower surfaces.
		Vertical Tail	– From front spar to trailing edge.
		Elevator	– The entire elevator external surface.
“II” (1)	-	Rudder	– The entire rudder external surface.
		WFF	– The entire wing-to-fuselage fairing (WFF) external surface.
			– The remaining external surfaces.

(1) The area “II” is called “non-critical”.

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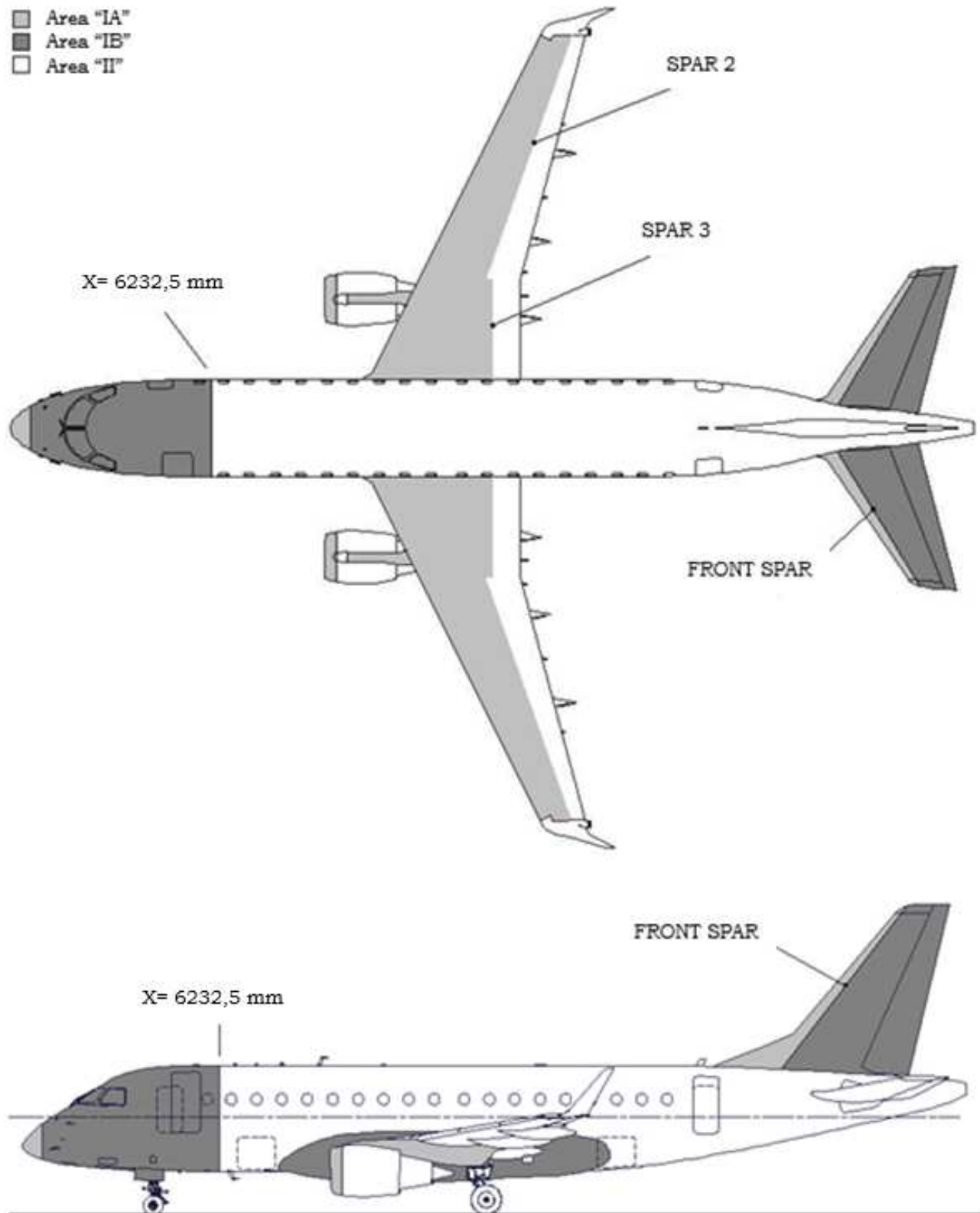


FIGURE 1 : AERODYNAMIC AREA CLASSIFICATION – ILLUSTRATION

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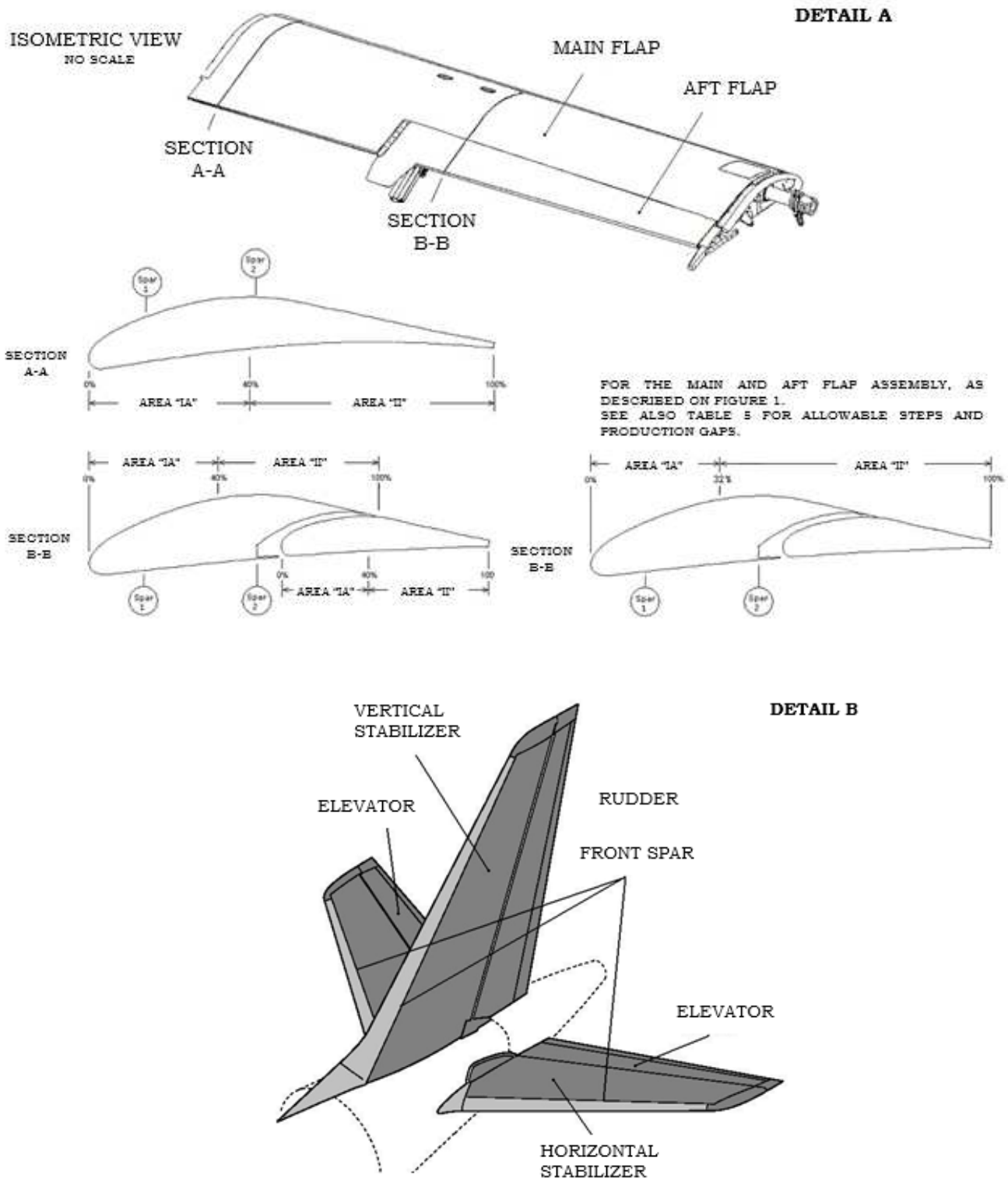


FIGURE 2 : AERODYNAMIC AREA CLASSIFICATION – DETAILS A AND B

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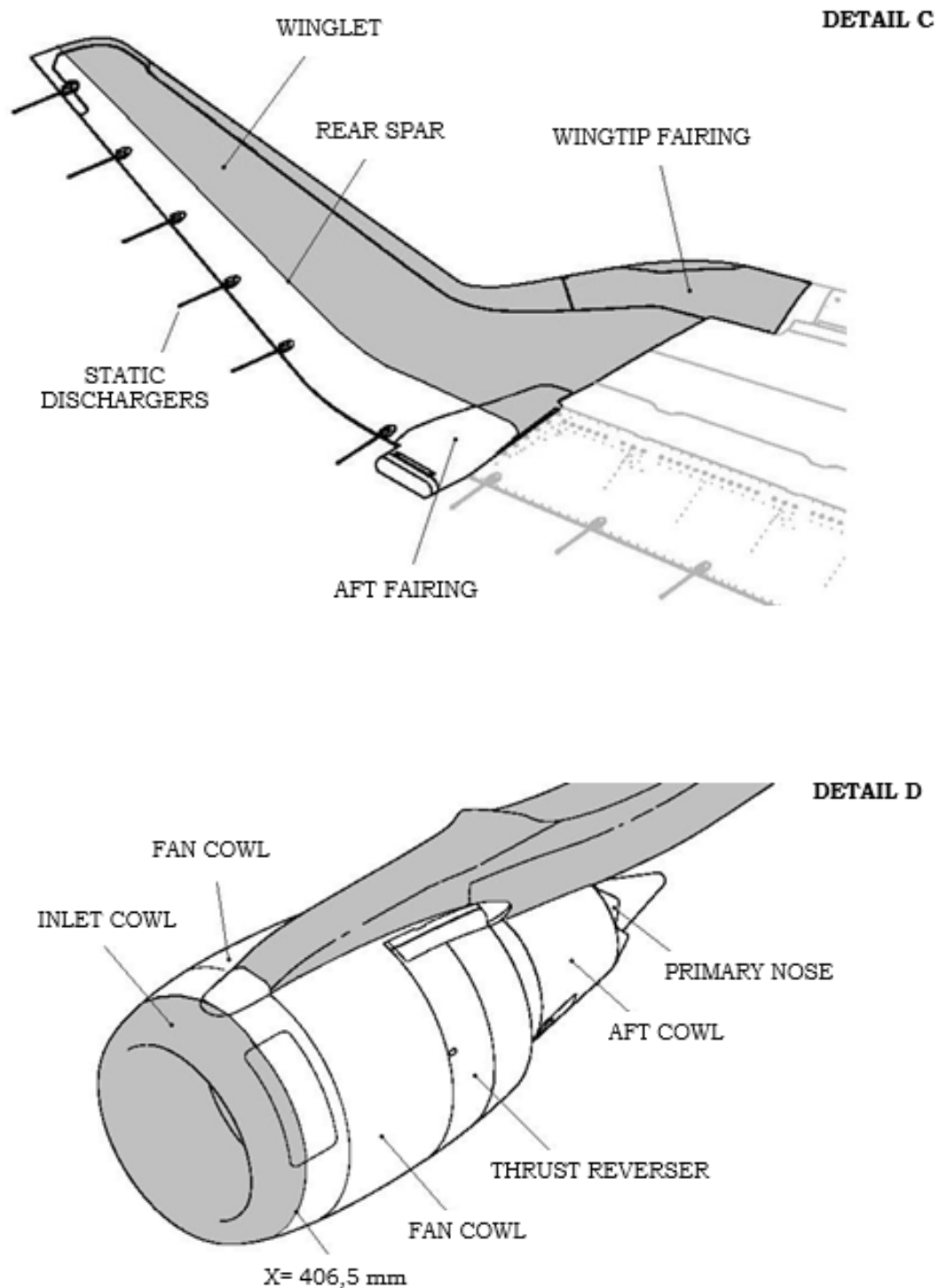


FIGURE 3 : AERODYNAMIC AREA CLASSIFICATION – DETAILS C AND D

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5. REQUIREMENTS

The requirements listed hereinafter are applicable to the areas "IA", "IB" and "II", unless otherwise specified. See Item 3.

5.1 SURFACE WAVINESS

Single, compound (total or partial) and periodic waves shown in Figure 4, Figure 5 and Figure 6 shall meet the aerodynamic smoothness requirements specified in Figure 7 and Figure 8. See Item 5.3.

NOTES

- 1) The wavelength shall be measured parallel to flight direction. Waves on riveting areas (normal to flight direction) only affect visual aspect. See NE 07-038.
- 2) For waves located at the edge of a part, where the measurement using two-point support for a straight-edge parallel to flight direction is not possible, the wavelength may be determined from consecutive spanwise measurements, starting from the edge, performed in different positions moving the measuring device chordwise until the wave height returns to zero in order to reconstruct the longitudinal wave. In such cases, the wave height (H) shall be the largest wave height measured and the wavelength (L) shall be the total spanwise measured length multiplied by two.

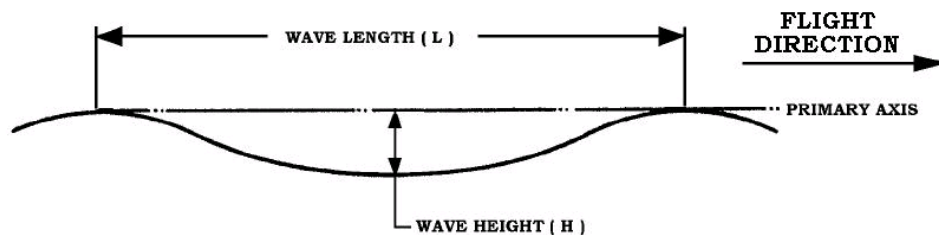


FIGURE 4 : WAVE TYPES – SINGLE WAVE

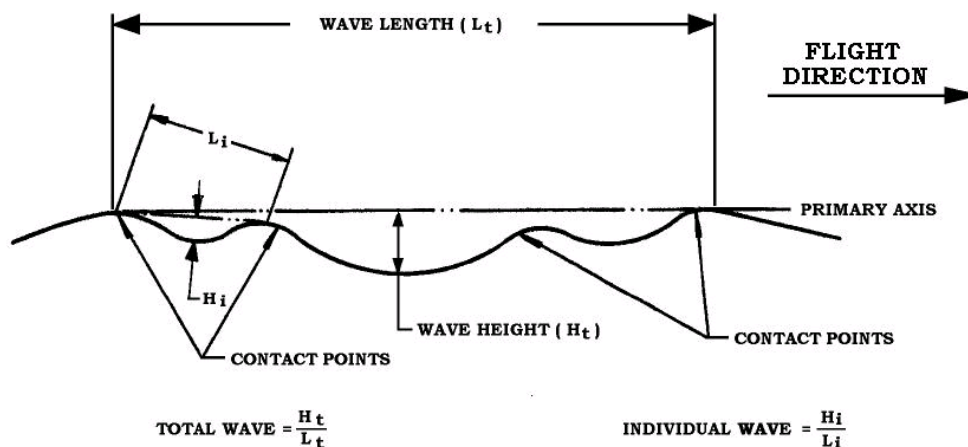


FIGURE 5 : WAVE TYPES – COMPOUND WAVE

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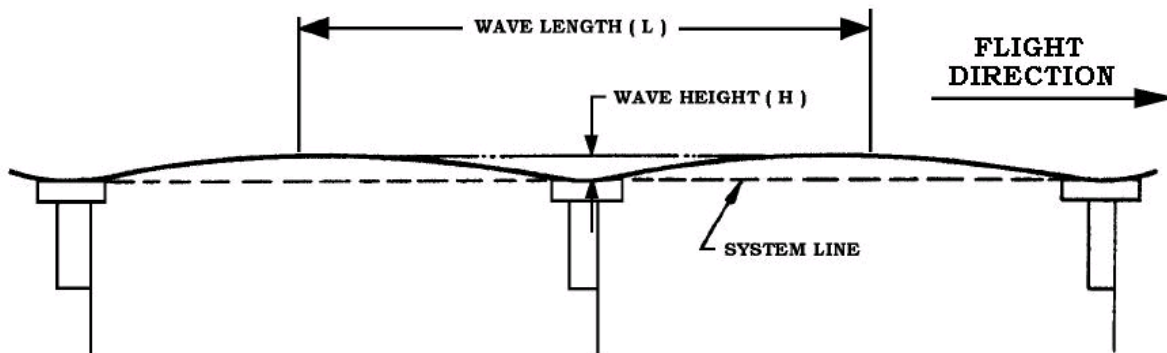
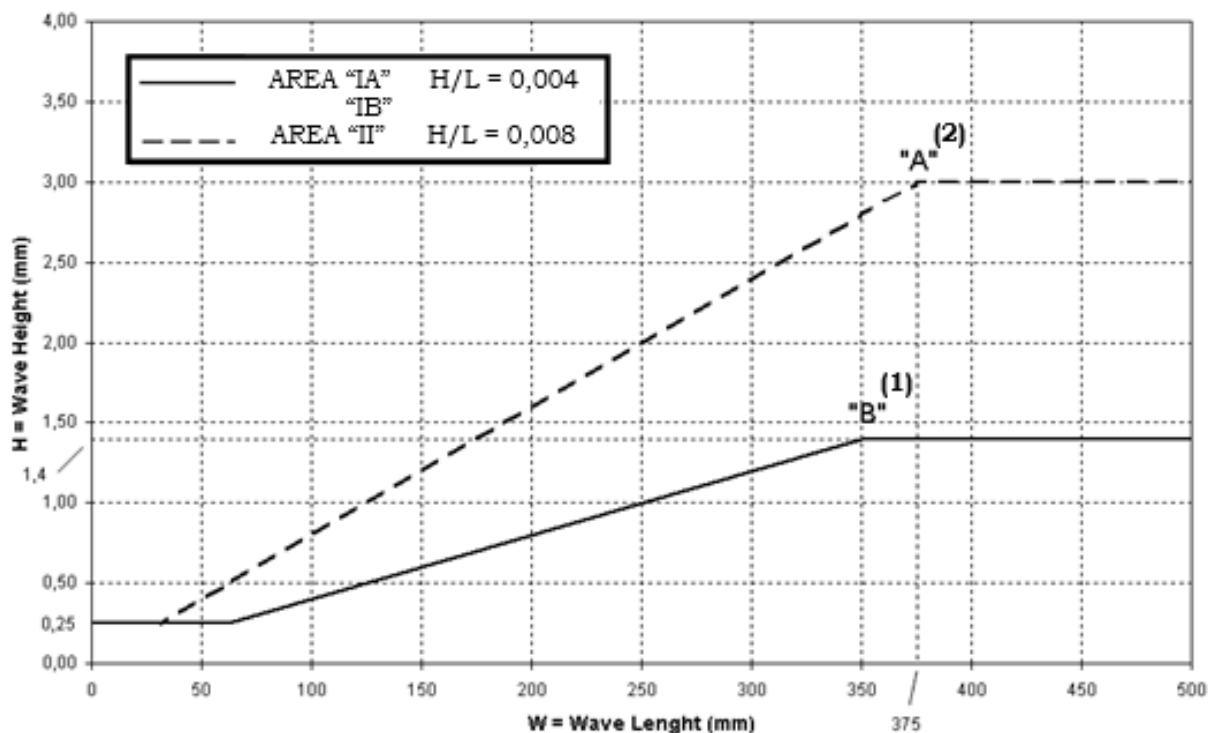


FIGURE 6 : WAVE TYPES – PERIODIC WAVE



- (1) Maximum wave height (H) value to area "IA" and "IB" is 1,4 mm for wavelength (W) greater than 350 mm.
 (2) Maximum wave height (H) value to area "II" is 3,0 mm for wavelength (W) greater than 375 mm.

Point "A" : $L_A = \frac{H_A}{0,008} = \frac{3}{0,008} = 375mm$

Point "B" : $H_B = L_B \times 0,004 = 350 \times 0,004 = 1,4mm$

FIGURE 7 : MAXIMUM ALLOWABLE HEIGHTS FOR WAVES

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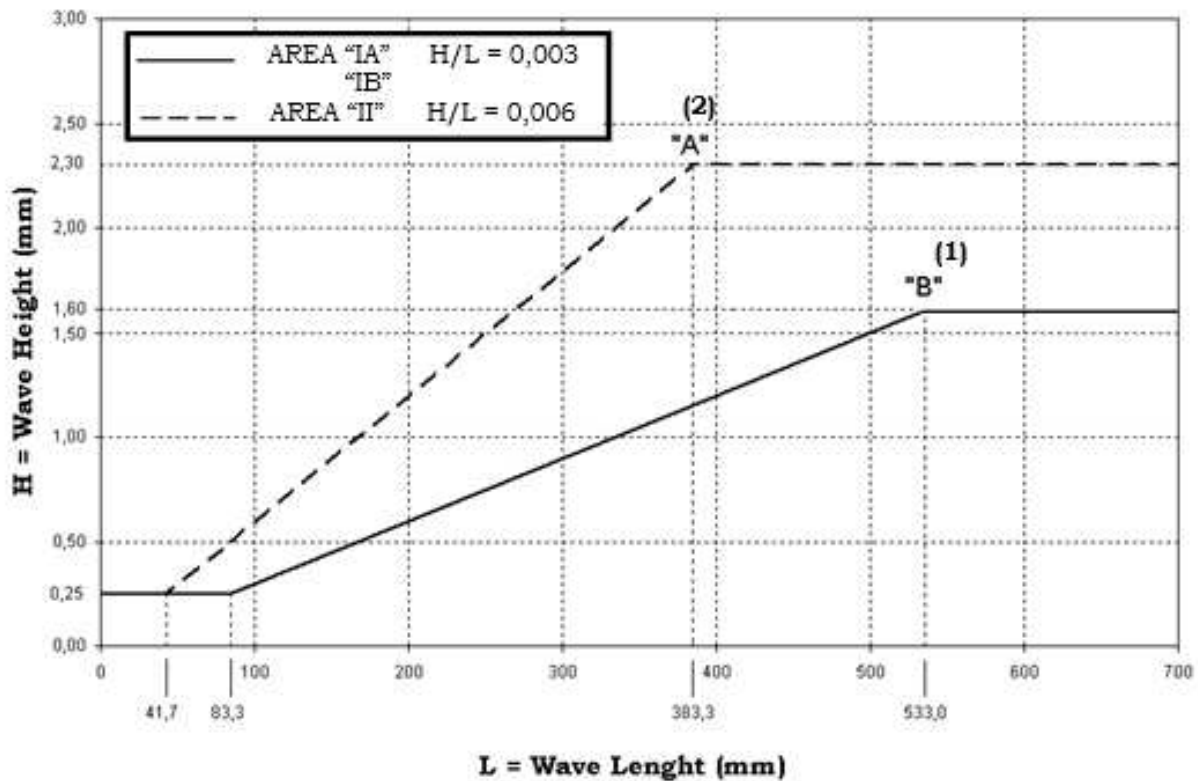
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NOTE

The Figure 8 establishes the maximum allowable heights for waves in nacelles, according to the reference in Item 8.1.



(1) Maximum wave height (H) value to area "IA" and "IB" is 1,6 mm for wavelength greater than 533 mm.

(2) Maximum wave height (H) value to area "II" is 2,3 mm for wavelength greater than 383,3 mm.

$$\text{Point "A": } L_A = \frac{H_A}{0,006} = \frac{2,3}{0,006} = 383mm$$

$$\text{Point "B": } L_A = \frac{H_A}{0,003} = \frac{1,6}{0,003} = 533mm$$

FIGURE 8 : MAXIMUM ALLOWABLE HEIGHTS FOR WAVES IN NACELLES

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5.2 SURFACE ROUGHNESS

The roughness of the aerodynamic surfaces is related to the depth of grooves and size of grains distributed along each surface, depending on the finishing process.

The surface finish roughness for external surfaces shall not exceed 3,2 μm (125 μin) Ra with a 2,50 mm (0,10 in) of sampling length (cutoff). For Ra and sampling length definitions. See NE 03-004 (or NE 03-085 for new drawings).

5.3 BASIC CONTOUR LINES

The aircraft contour lines deviations shall be within the tolerances indicated in Table 2 and Figure 9. The requirements of both Item 5.1 and Item 5.3 shall be met.

TABLE 2 : BASIC CONTOUR LINE TOLERANCE

AREA (1)	TOLERANCE (mm)
"IA"	1,4
"IB"	2,2
"II"	3,0

(1) The contour line deviation tolerances at the area "IA" and "II" of the nacelles are 1,6 mm (0,032 in) and 3,2 mm (0,126 in), respectively.

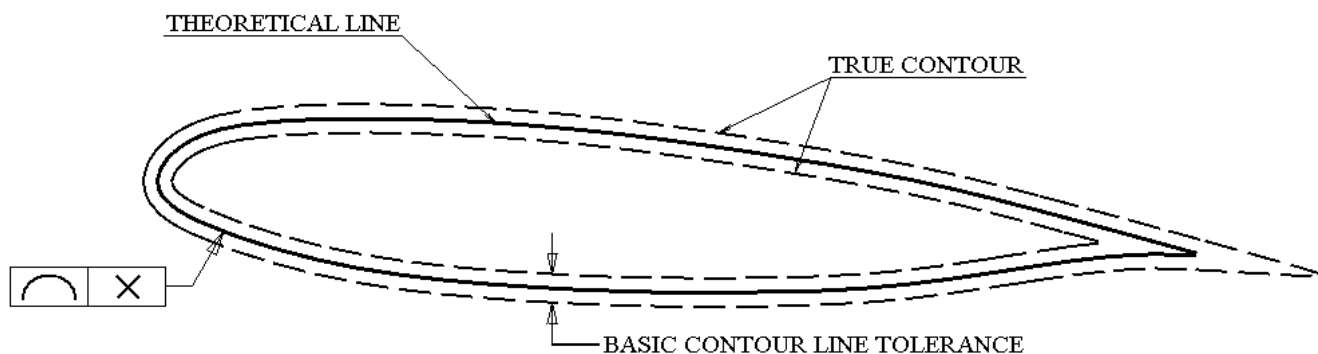


FIGURE 9 : BASIC CONTOUR LINE TOLERANCE

The contour line tolerance requirement is specified for the entire span of aircraft components; however, specific control sections shall be defined in agreement with the Embraer Aerodynamics Engineering Group.

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5.4 STEPS AND GAPS

Multiple steps and gaps are present on the external surface on an aircraft. The influence of these discrepancies on aerodynamic performance may be markedly strong, thus the need for controlling them.

The definition of normal and parallel steps and gaps is shown in Figure 10. Gaps and steps between the control surfaces (ailerons, speed brake and ground spoilers) and the adjacent parts between the doors in pressurized areas and the adjacent parts are discussed separately as special cases.

NOTE

The sign convention for normal steps is presented in Figure 11 are:

"+" means steps "against" the flight direction; and

"-" means "favorable" steps.

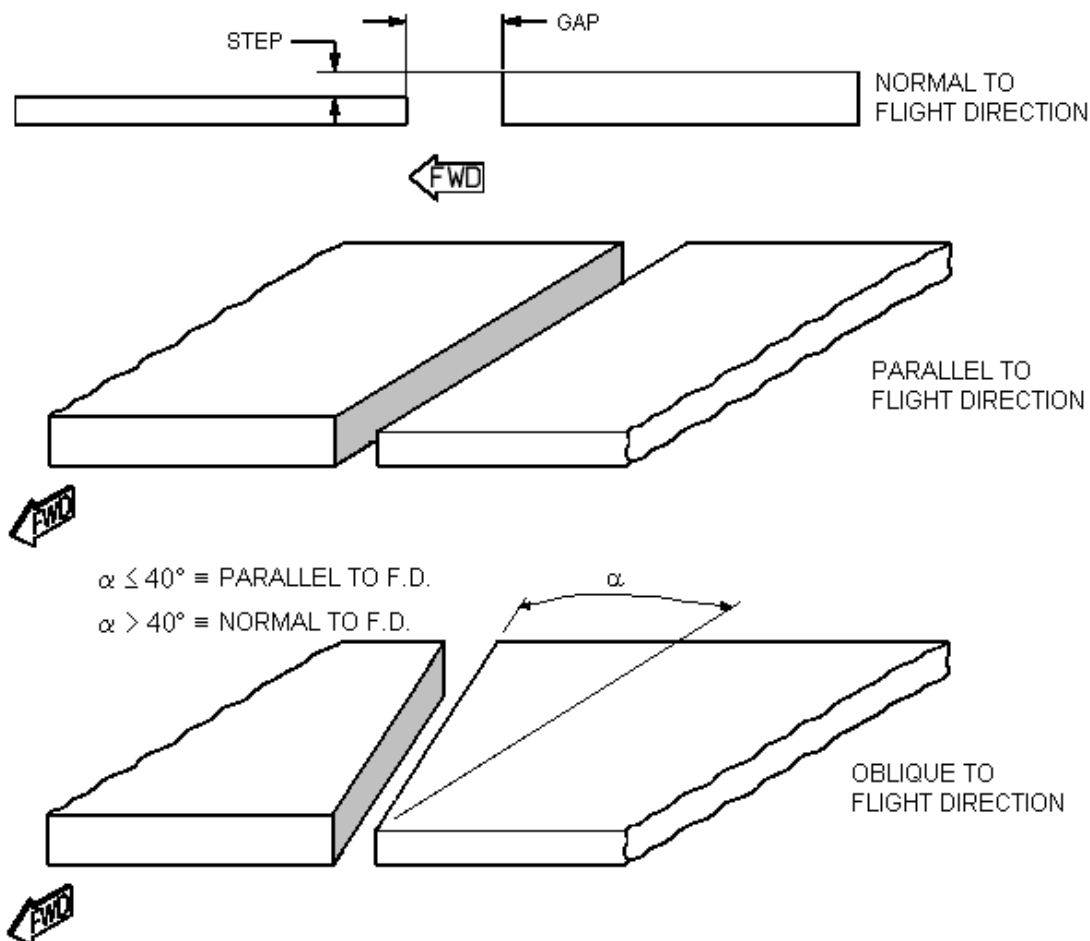


FIGURE 10 : STEPS AND GAPS DEFINITIONS

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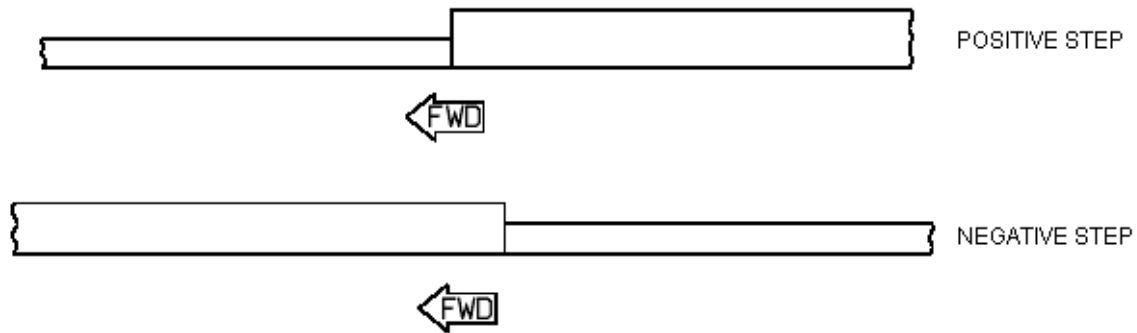


FIGURE 11 : SIGN CONVENTION FOR NORMAL STEPS

5.4.1 Fasteners

- a) protuberant head fasteners shall not be used on area “IA” and “IB”. The use of protuberant head fasteners on area “II” shall be subject of analysis and approval of Embraer Aerodynamics Group on a case-by-case basis;
- b) countersunk head fasteners shall be according to Figure 12.

NOTES

- 1) Fasteners not complying with aerodynamic smoothness tolerances herein specified may be accepted provided that they do not exceed 10% of the fasteners that are installed in the airplane and that they are not located in areas “IA” and “IB”.
- 2) The maximum values for upper and lower deviations shall be according to NE 03-006 for rivets and pins or NE 03-068 for bolts and screws.

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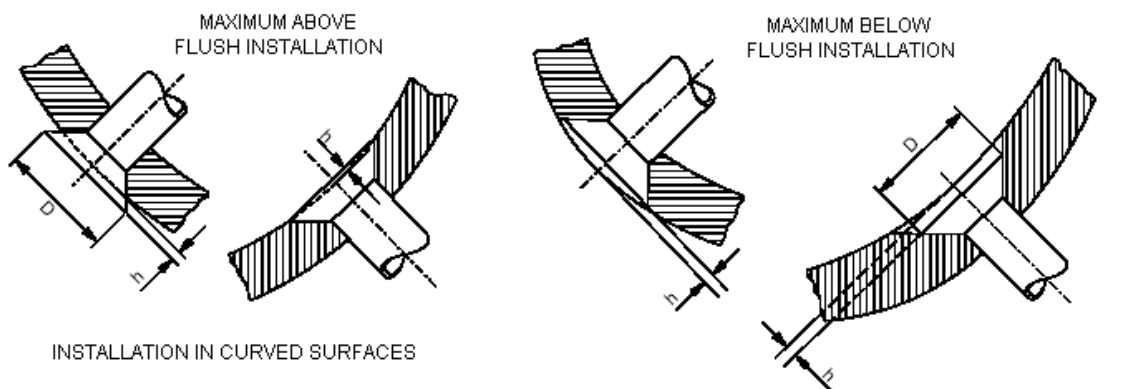
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COUNTERSUNK FASTENER



h VALUE MUST BE WITHIN THE FOLLOWING LIMITS:

- (I) HEAD ABOVE SHEET COUNTERSUNK OR DIMPLED:
 $h \leq 0,020.D$
- (II) HEAD BELOW SHEET COUNTERSUNK OR DIMPLED:
 $h \leq 0,030.D$

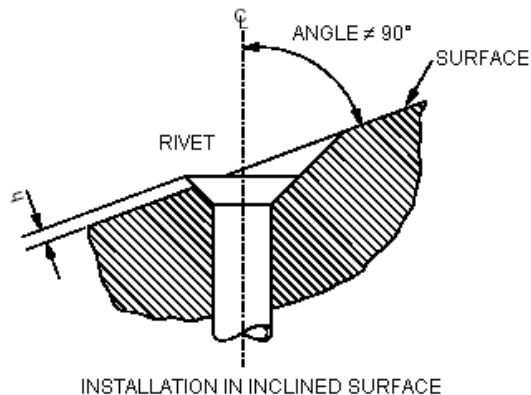
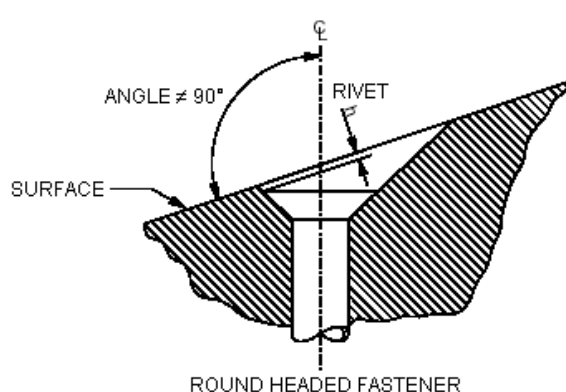
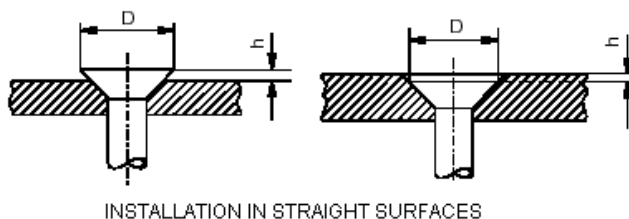


FIGURE 12 : ALLOWABLE TOLERANCES FOR COUNTERSUNK HEAD FASTENER POSITION IN REGARD TO SHEET SURFACE

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5.4.2 Skin Joints, Access Panels and Doors

The maximum allowable steps and gaps for joints (for example: plate splices, access doors, inspection doors, etc.) shall be according to Table 3. Allowable steps and gaps for doors in pressurized areas are given in Item 5.4.3. Definition of steps and gaps is given in Figure 13.

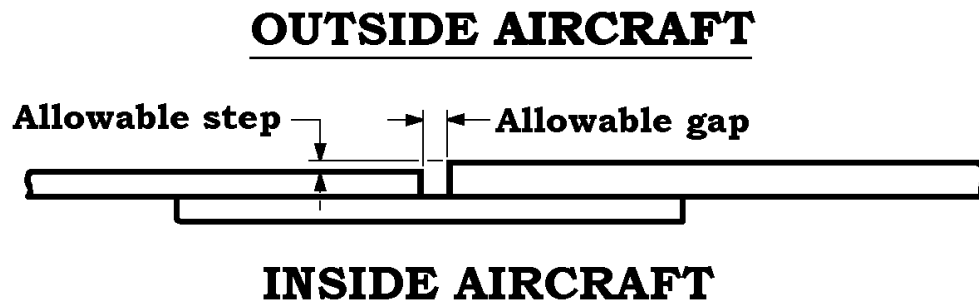


FIGURE 13 : ALLOWABLE TYPE OF SKIN JOINTS BETWEEN FIXED SURFACES

TABLE 3 : MAXIMUM ALLOWABLE STEPS AND GAPS FOR SKIN JOINTS

JOINT TYPE	AREA	STEP (mm)		GAP (mm) (3)	
		NORMAL TO F.D. (1)	PARALLEL TO F.D. (1)	NORMAL TO F.D. (1)	PARALLEL TO F.D. (1)
FIXED SURFACES (5) (8)	"IA"	±0,5	0,5	1,6	1,6
	"IB"	±0,5	0,5 (6)	1,6	1,6
	"II"	±0,8	0,8	2,0	2,0
SKIN AND BOLTED ACCESS DOORS AND INSPECTION PLATES	"IA"	±0,5	0,5	2,0	2,0
	"IB"	±0,5	0,5 (6)	2,0	2,0
	"II"	±0,8	0,8	2,0	2,0
SKIN AND ALL OTHER KINDS OF ACCESS DOORS OR PLATES	"IA"	±1,0	1,0	2,0	2,0
	"IB"	±1,0	1,0	2,0	2,0
	"II"	±1,0	1,0	3,0	3,0
NACELLE MAJOR COMPONENTS JOINTS	"IA"	±0,4	0,4	2,0	2,0
	"II"	±0,6 (9)	0,6	2,0 (10)	2,0
NACELLE RIVETED, WELDED AND BOLTED JOINTS	"IA"	±0,4	0,4	2,0	2,0
	"II"	±0,6	0,6	2,0	2,0
NOSE LANDING GEAR DOORS (4)	"IB"	±1,0	1,0	5,0	5,0
MAIN LANDING GEAR DOORS (4)	"IA"	±1,0	1,0	5,0	5,0
WING-TO-FUSELAGE FAIRING PANELS	"IB"	±0,8	0,8	6,5 (11)	6,5 (11)
RADOME AND FUSELAGE NOSE	"IB"	±1,1 (7)	N.A. (2)	2,5	N.A. (2)

(1) F.D. = Flight Direction.

(2) N.A. = Not Applicable.

The notes of Table 3 continue.

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The notes of Table 3 continued.

(3) All gaps between fixed surfaces shall be filled with sealant and smoothed. The maximum allowable ridge or depression between sealant and skin surface shall be according to Figure 14.

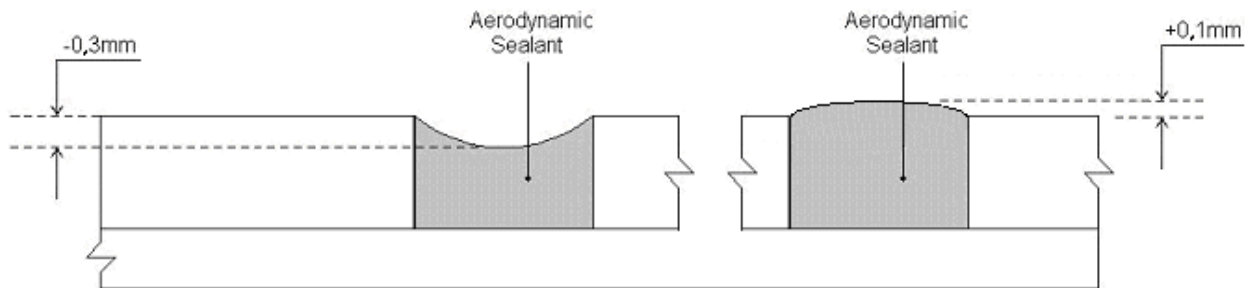


FIGURE 14 : MAXIMUM ALLOWABLE SEALANT RIDGE OR DEPRESSION

- (4)** The landing gear door gaps shall be sealed in the retracted position.
- (5)** The gap between the skin and the removable leading edge of the empennages shall be sealed and smoothed after each removal and installation.
- (6)** The tolerance for steps parallel to F.D. at horizontal and vertical stabilizers is 0,8 mm.
- (7)** Radome steps of up to 1,4 mm are allowed up to 20% of its perimeter.
- (8)** Due to drag considerations, the use of lap joints shall be avoided, and it only may be implemented after Aerodynamics Department approval. The allowable steps for the lap joint areas will be the thickness of the skin plus the tolerance for fixed joints in each surface area ("IA", "IB" and "II").
- (9)** The maximum step between fan cowl and thrust reverser is $\pm 0,4$ mm.
- (10)** The maximum gap between inlet and fan cowls is 4,0 mm.
- (11)** The maximum gap (normal and parallel to the airflow) for reference only. This gap is controlled during panel assembly by measuring the gap between access panels and wing-to-fuselage fairings (forward, middle and rear) at the bottom area as it is shown in Figure 15.

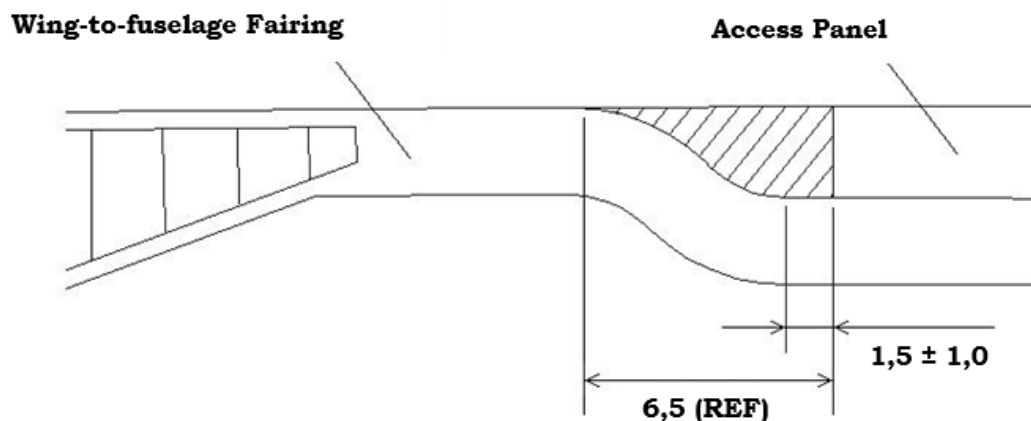


FIGURE 15 : GAP TOLERANCE FOR THE WING-TO-FUSELAGE FAIRING ACCESS PANELS

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5.4.3 Doors and Windows in Pressurized Areas

The maximum allowable steps and gaps for doors and windows located in pressurized areas shall be according to Table 4.

TABLE 4 : MAXIMUM ALLOWABLE STEPS AND GAPS FOR DOORS AND WINDOWS IN PRESSURIZED AREAS (2)

ITEM	ALLOWABLE STEP (mm)		ALLOWABLE GAP (mm)
	INSIDE	OUTSIDE	
MAIN AND SERVICE DOORS	1,5	1,5	4,0
BAGGAGE DOORS (5)	2,0 (4)	2,0 (4)	4,0
WINDSHIELD (1)	0,5	0,5	4,3
MOVEABLE COCKPIT WINDOWS	1,0	1,0	4,5 (3)
PASSENGER WINDOWS (1)	1,0	1,0	2,0

(1) Gaps shall be filled in with sealant.

(2) The above tolerances are for the aircraft pressurized.

(3) Movable cockpit windows gaps of up to 5,5 mm are allowed up to 20% of its perimeter.

(4) Baggage doors steps of up to $\pm 2,5$ mm are allowed up to 20% of its perimeter.

(5) The above specified step tolerances are for the baggage doors are not applicable at the bottom of the door if the scuff plate is installed. In this case, step control in this area is not required.

5.4.4 Control Surfaces, Flaps and Slats

The allowable steps and gaps and allowable deviations shall be as follows:

a) due to deviations in the manufacturing and assembly processes of the aircraft parts, gaps and steps not predicted during design may be found and shall be controlled, along the entire interfaces, in order to minimize the aerodynamic performance degradation.

Table 5 presents the tolerance for steps and gaps between retracted control surfaces and fixed parts;

TABLE 5 : ALLOWABLE STEPS AND PRODUCTION GAPS BETWEEN CONTROL SURFACES, RETRACTED FLAPS AND SLATS AND FIXED PARTS (4) (5) (6) (7) (8)

SURFACE	STEP (mm)		GAP (mm)	
	NORMAL TO F.D. (1)	PARALLEL TO F.D. (1)	NORMAL TO F.D. (1)	PARALLEL TO F.D. (1)
RETRACTED SLAT TO WING	$\pm 1,5$	2,0	N.A. (2)	(3)
AILERON TO WING	$\pm 1,5$	1,5	N.A. (2)	(3)
SPOILERS TO WING	$\pm 1,5$	1,5	(3)	(3)
RETRACTED FLAP TO LOWER SHROUD	$\pm 1,5$	1,5	2,0	5,0
RETRACTED FLAP TO SPOILERS	$\pm 1,5$	1,5	N.A. (2)	(3)

The Table 5 continue.

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The Table 5 continue.

TABLE 5 : ALLOWABLE STEPS AND PRODUCTION GAPS BETWEEN CONTROL SURFACES, RETRACTED FLAPS AND SLATS AND FIXED PARTS (4) (5) (6) (7) (8)

SURFACE	STEP (mm)		GAP (mm)	
	NORMAL TO F.D. (1)	PARALLEL TO F.D. (1)	NORMAL TO F.D. (1)	PARALLEL TO F.D. (1)
RETRACTED MAIN FLAP TO AFT FLAP	±1,5	1,5	N.A. (2)	7,0
RETRACTED FLAP TO TORQUE BOX	N.A. (2)	1,5	N.A. (2)	(3)
RETRACTED FLAP TO FLAP TRACK FAIRING	±1,5	N.A. (2)	N.A. (2)	N.A. (2)
ELEVATOR TO HORIZONTAL STABILIZER	±1,5	1,5	N.A. (2)	5,0
RUDDER TO VERTICAL STABILIZER	±1,5	1,5	N.A. (2)	5,0

(1) F.D. = Flight Direction.

(2) N.A. = Not Applicable.

(3) The tolerances for gaps normal or parallel to the airflow are specified in the respective installation drawings once these tolerances change from one gap to another.

(4) Longitudinal and lateral gaps between the movable surfaces and other parts shall be aerodynamically sealed when the surface is in the nominal position.

(5) Longitudinal gap between the inboard and outboard flaps shall be sealed for all slat deflections.

(6) Longitudinal gap between slat panels shall be sealed for all slat deflections.

(7) In areas with special gaps and step requirements (for instance, hinge point areas), the tolerance shall be specified in applicable drawings.

(8) The step between the control surface or flap and the aircraft fixed parts shall be measured according to Figure 16. The step is considered zero when the extended contour line of the fixed part matches exactly that of the control surface. The sign convention for the steps is defined in Figure 11.

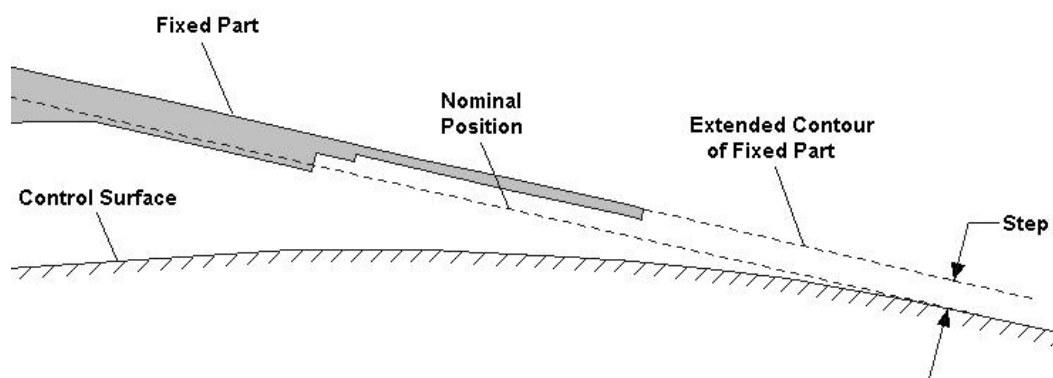


FIGURE 16 : STEPS BETWEEN CONTROL SURFACES AND FIXED PARTS

b) some surfaces (such as deflected flaps and slats) require a large but well-defined gaps between them and the aircraft fixed parts. The purpose of these gaps is related to the aerodynamic performance of the surface. Table 6 presents the tolerance for gaps between deflected control surfaces and fixed parts.

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**TABLE 6 : ALLOWABLE DEVIATIONS IN THE AERODYNAMIC GAPS BETWEEN
DEFLECTED FLAPS , SLATS AND FIXED PARTS (1)**

SURFACE	GAP (mm)	OVERLAP (mm)
SLATS AND WING (2)	±1	±2
SLATS AND WING (3)	±2	±2
INBOARD FLAP MAIN ELEMENT AND WING	±5	±5
INBOARD FLAP MAIN AND AFT ELEMENTS	±2	±2
OUTBOARD FLAP MAIN ELEMENT AND WING	±5	±5
OUTBOARD FLAP MAIN AND AFT ELEMENTS	±2	±2

- (1) The tolerances for gaps shall be observed for the entire range of surface deflection.
(2) Reference Slat Angle = 15°.
(3) Reference Slat Angle = 25°.

5.5 CONTROL SURFACES, SLAT AND FLAP ANGULAR DEFLECTION TOLERANCES

The control surface deflection angle deviations may be divided into two types:

- Absolute Deviation: Angular deviation between the actual (measured) deflection and the nominal one.
- Symmetry Deviation: Difference between the actual (measured) deflections of the port and starboard parts of the same control surface, unless otherwise stated.

The maximum allowable absolute and symmetry deviations for control surface and flap deflections are given in Table 7.

These values are applicable to the surfaces either at their neutral positions or at their maximum deflections, unless otherwise stated.

Both absolute and symmetrical deviations have to be respected at the same time.

**TABLE 7 : MAXIMUM ALLOWABLE ABSOLUTE AND SYMMETRY DEVIATIONS FOR
CONTROL SURFACE, SLAT AND FLAP DEFLECTIONS**

SURFACE	ABSOLUTE DEVIATION (DEG)	SYMMETRY DEVIATION (DEG)
AILERONS (2)	±0,5	±1,0
SPOILERS	±1,0	±1,0
HORIZONTAL STABILIZER	±0,5	±0,5
ELEVATORS	±0,5	±1,0
RUDDER	±1,0	N.A. (1)
TAKE-OFF SLATS	±0,5	±1,0
LANDING SLATS	±1,0	±1,0
RETRACTED FLAPS	±0,5	±1,0
TAKE-OFF FLAPS, MAIN ELEMENT, INBOARD AND OUTBOARD	±1,0	±1,0

The Table 7 continue.

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The Table 7 continue.

TABLE 7 : MAXIMUM ALLOWABLE ABSOLUTE AND SYMMETRY DEVIATIONS FOR CONTROL SURFACE, SLAT AND FLAP DEFLECTIONS

SURFACE	ABSOLUTE DEVIATION (DEG)	SYMMETRY DEVIATION (DEG)
TAKE-OFF FLAPS, AFT ELEMENT, INBOARD AND OUTBOARD	±1,0	±1,0
LANDING FLAPS, MAIN ELEMENT, INBOARD AND OUTBOARD	±1,5	±1,0
LANDING FLAPS, AFT ELEMENT, INBOARD AND OUTBOARD	±1,5	±1,0
VENTRAL SPEED BRAKE (IF APPLICABLE)	±2,0	N.A. (1)

(1) N.A.: Not Applicable.

(2) The maximum allowable absolute and symmetry deviations for the aileron at the neutral position are, respectively: ±0,2° and ±0,4°.

5.6 COMPONENT ANGULAR TOLERANCES

The angular deviations of aircraft components may be divided into two types:

- Absolute Deviation: Angular deviation between the actual and nominal position.
- Symmetry or Inter-Component Deviation: Angular difference between symmetrical stations (for example: port and starboard wing, pylons, horizontal tail) or between components of the same part (for example: front, center and rear fuselage).

The component tolerances are given in Table 8.

TABLE 8 : MAXIMUM ALLOWABLE COMPONENT ANGULAR DEVIATIONS

AIRCRAFT COMPONENT	MEASURED ANGLE	ABSOLUTE DEVIATION (minute)	SYMMETRY OR INTER-COMPONENT DEVIATION (minute)
WING	INCIDENCE AT ROOT	±10	±20
	ROOT-TIP TWIST	±10	±20
	DIHEDRAL	±20	±20
	SWEEP	±10	±10
WINGLET	INCIDENCE	±30	±30
	DIHEDRAL	±60	±60
	SWEEP	±30	±30
HORIZONTAL STABILIZER	INCIDENCE AT ROOT	±10	±20
	ROOT-TIP TWIST	±10	±20
	DIHEDRAL	±20	±20
	SWEEP	±10	±10
VERTICAL TAIL	LATERAL INCIDENCE AT ROOT	± 10	N.A. (1)
	ROOT-TIP TWIST	± 10	N.A. (1)
	VERTICAL ALIGNMENT	± 15	N.A. (1)
	SWEEP	± 20	N.A. (1)

The Table 8 continue.

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The Table 8 continue.

TABLE 8 : MAXIMUM ALLOWABLE COMPONENT ANGULAR DEVIATIONS

AIRCRAFT COMPONENT	MEASURED ANGLE	ABSOLUTE DEVIATION (minute)	SYMMETRY OR INTER-COMPONENT DEVIATION (minute)
PYLONS	INCIDENCE (ELEVATION)	±10	±10
	INCIDENCE (AZIMUTH)	±10	±10
	VERTICAL ALIGNMENT	±20	±20
FUSELAGE	ALIGNMENT OF SECTIONS WITH LONGITUDINAL AXIS	±10	±10
ENGINE	LONGITUDINAL ALIGNMENT	±10	±10
	LATERAL ALIGNMENT	±10	±10
BLADE ANTENNAS	NOMINAL INCIDENCE	±60	N.A. (1)

(1) N.A. = Not Applicable.

5.7 SPECIAL AERODYNAMIC REQUIREMENTS

Special aerodynamic smoothness requirements or exceptions to the values on this Standard shall be specified on the production drawings, after analysis and approval by the Embraer Aerodynamic Group.

6. QUALITY ASSURANCE PROVISIONS

6.1 The individual responsible for performing aerodynamic tolerance compliance shall be trained and shall guarantee the fulfillment of the requirements of this Standard.

6.2 Embraer Quality Assurance shall monitor the process of aerodynamic surface fabrication by an auditing process so as to guarantee the fulfillment of Item 6.1 and promote preventive actions when necessary.

7. GENERAL NOTES

7.1 DRAWING CALLOUT

This Standard shall be called out on the skin assembly drawings for the fuselage, wing, horizontal and vertical tails and pylons fairing, as well as on the drawings related to the entire assemblies, through the drawing note **below**:

NI 884 - Aerodynamic Smoothness, according to NE 02-044.

8. BIBLIOGRAPHY

This Standard has been based on the following documents:

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8.1 Appendix IIC: Nacelle Aerodynamic Smoothness Requirements of Section 2.
Power-plant Specification of Document CF34-8E Power-plant System for Embraer ERJ
170/ERJ 190 - Exhibit A1-1 Contractors Technical Proposal.

8.2 NE 02-041: Aerodynamic Smoothness - Requirements for EMB-145.

8.3 314-EAD-VA97-01: External Surface Aerodynamic Smoothness and Angular
Tolerance Requirements for the EMB-314 Aircraft.

8.4 170-GEA-VA99-01: External Surface Aerodynamic Smoothness and Angular
Tolerance Requirements for ERJ 170 Aircraft.

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